AN AGENTIC MULTIMODAL AI FRAMEWORK FOR AIRCRAFT TECHLOG INTELLIGENCE USING RETRIEVAL-AUGMENTED GENERATION, PREDICTIVE MODELING, AND SAFETY-CONSTRAINED DECISION SUPPORT

AIMLCZG628T: Dissertation / Project Work

by

Ankit Bhardwaj
2024AA05939

Dissertation work carried out at

The Emirates Group, Dubai, UAE

Submitted in partial fulfilment of
Master of Technology (Artificial Intelligence & Machine Learning)
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Under the Supervision of

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BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE
PILANI (RAJASTHAN)

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ABSTRACT

Aircraft maintenance operations at large carriers generate thousands of defect entries every month through electronic technical logbooks. Engineers working under time pressure must cross-reference maintenance manuals, MEL/CDL provisions, engineering orders, and prior resolved cases -- often pulling from systems that do not talk to each other. The cognitive load is significant, and the cost of a missed or delayed defect resolution is high both in aircraft-on-ground time and safety exposure.

This dissertation develops an Agentic Multimodal AI Framework -- referred to throughout this report as the Techlog Copilot -- that assists maintenance engineers by triaging defects across eight fault categories, retrieving cited evidence from a curated knowledge corpus, predicting recurring defects on a per-tail basis, and generating explainable recommendations within safety governance constraints aligned to EASA Level 1/2 guidance.

At the mid-semester stage, the foundational data acquisition and preparation pipeline has been completed. This covers the download and normalisation of four real public aviation datasets, the generation of a schema-consistent synthetic corpus for model training, and the construction of chronological train/val/test splits for both the triage and recurrence tasks. The next phase focuses on implementing the RAG pipeline, training the triage and recurrence models, and integrating the agent orchestration layer.

Signature of the Student: _______________ Date: _______________

Name: Ankit Bhardwaj

Signature of the Supervisor: ____________ Date: _______________

Name: ___________________________

Place: Dubai, UAE

CONTENTS

1. INTRODUCTION AND PROBLEM CONTEXT

The aircraft technical logbook is one of the most operationally critical documents in commercial aviation. Every defect a pilot reports, every maintenance action an engineer carries out, and every deferred item under a MEL provision gets recorded there. At a large hub carrier operating several hundred aircraft, the volume of these entries runs into the thousands per week. Manually searching through this data to identify patterns, retrieve the right manual section, or catch a recurring fault before it escalates is both time-consuming and error-prone.

The practical problem addressed by this dissertation is straightforward to state but harder to solve: how do you build a system that an engineer can query in natural language, get a relevant evidence-backed recommendation from, and trust enough to act on -- all within the three-to-five minute turnaround window that line maintenance typically allows?

Existing maintenance support tools at most carriers are largely retrieval-based. Engineers search PDF manuals by keyword, look up MEL items by ATA chapter, and rely on institutional memory for recurring fault patterns. There is little integration between these sources and no predictive layer. The research opportunity is to combine recent progress in retrieval-augmented generation, multimodal deep learning, and agentic AI frameworks into something that fits the operational constraints of an airline MRO environment.

The proposed framework consists of five interacting components: a retrieval agent drawing on a knowledge corpus of 1,213 documents; a triage agent classifying defects across eight ATA-aligned fault categories; a recurrence prediction agent operating on per-tail defect histories; a reasoning agent using a parameter-efficient fine-tuned LLM; and a safety layer handling confidence gating, abstention, and audit logging. The safety governance approach is aligned to EASA's Concept Paper on Level 1 and Level 2 Machine Learning Applications (Issue 2, 2023).

The dissertation is intended as a research prototype for academic purposes. No proprietary airline data is used at any stage; all real data comes from publicly accessible sources, and the synthetic component is clearly separated throughout.

2. SYSTEM ARCHITECTURE

The framework is built as a multi-agent system orchestrated through LangGraph, with each agent responsible for a distinct function in the maintenance decision pipeline. Figure 1 below shows the high-level dataflow.

Figure 1: Techlog Copilot -- Multi-Agent System Architecture

Triage Agent

A multimodal MLP taking three input streams: the defect narrative text embedded via a sentence-transformer, structured metadata (ATA chapter, flight phase, aircraft type, severity), and 42 C-MAPSS-style sensor features (last value and slope per sensor). The eight output classes map to the fault taxonomy derived from ATA iSpec 2200 chapter groupings.

Recurrence Agent

An LSTM encoder operating on a chronologically ordered sequence of prior defect events per tail number. Input features include defect category, ATA chapter, severity, and days elapsed since the last event. The binary output predicts whether the same defect category will recur within a 30-day horizon.

Retrieval Agent

A RAG pipeline using Chroma as the vector store and sentence-transformers for document embedding. The knowledge corpus at this stage contains 1,213 documents spanning real FAA Airworthiness Directives, synthetic AMM/MEL/EO/ADV procedures, and de-identified case narratives from NASA ASRS.

Reasoning Agent

Mistral-7B with LoRA adapters fine-tuned on the 26,210 real defect-to-action pairs extracted from FAA Service Difficulty Reports. The model generates a ranked, cited recommendation given the retrieval context and triage output.

Safety Layer

A confidence gate that suppresses low-confidence outputs and routes them to a human escalation flag. Nine safety-critical question types in the QA benchmark are marked MUST_ABSTAIN; the system declines answering these rather than guessing. Every decision, including abstentions, is written to an immutable audit log.

3. DATA ACQUISITION AND PREPARATION -- WORK DONE

The first major deliverable of the dissertation is a reproducible data preparation pipeline (src/data/) that downloads, normalises, and generates all datasets required for the modelling tasks. The pipeline is deterministic (seed=42) and runs end-to-end with a single command: "python -m src.data.prepare_all".

3.1 Real Public Data

NASA C-MAPSS Turbofan Degradation Dataset

The four sub-datasets (FD001-FD004) from the NASA Prognostics Center of Excellence were downloaded and extracted. These files contain multivariate run-to-failure time series for turbofan engines across varying operating conditions. The schema -- three operational settings plus 21 sensor measurements -- is used directly in this project for building the signal-modality input to the triage classifier. Separate train, test, and RUL files are available for each sub-dataset.

FAA Service Difficulty Reports (SDR), 2023-2025

Three annual CSV exports were downloaded from the FAA's public endpoint, covering approximately 197,000 individual defect reports filed by US-registry operators. Each record follows a 76-column schema including JASC/ATA code, part name, condition, stage of operation, and a free-text Discrepancy field containing both the defect description and the corrective action. The normalisation step produced 196,118 normalised records, of which 166,037 were mapped to one of the eight taxonomy classes. The corrective action extraction produced 26,210 defect-to-action pairs used for fine-tuning the reasoning agent.

NASA ASRS Report Sets

Three curated report-set PDFs were downloaded from NASA's Aviation Safety Reporting System -- covering maintenance technician reports, fuel management issues, and cabin fumes/smoke incidents. A custom PDF parser was built using the pypdf library to segment reports by ACN marker, extract structured metadata fields, and parse free-text Narrative sections. This produced 150 structured records added to the knowledge corpus as CASE-type documents.

FAA Airworthiness Directives

500 AD rules were retrieved via the Federal Register API, with full text for 49 fetched from govinfo.gov. All 500 AD documents are included in the RAG corpus.

3.2 Synthetic Data

4,425 synthetic techlog records were generated across 70 tail numbers (A320, A321, A330, A350, B737, B777, B787 types). 563 knowledge corpus documents were generated across six types (AMM, MEL, AD, EO, ADV, CASE). 3,915 C-MAPSS-style signal windows with defect-specific degradation signatures were produced. A 189-item QA benchmark with MUST_CITE / CAN_ANSWER / MUST_ABSTAIN safety taxonomy was generated. The full knowledge corpus, after appending real FAA ADs and ASRS reports, totals 1,213 documents.

3.3 Train/Val/Test Splits

Splits were generated chronologically to prevent temporal leakage. Train/val/test fractions are 80/10/10. Triage: 3,540 / 442 / 443. Recurrence: 3,470 / 372 / 373. Positive recurrence rate in the training set is approximately 37%.

4. DATASET DESCRIPTION AND STATISTICS

Dataset	Type	Path	Volume
NASA C-MAPSS FD001-FD004	Real	data/raw/nasa_cmapss/	14 files (train/test/RUL)
FAA SDR 2023-2025	Real	data/raw/faa_sdr/real/	~197,000 rows, 76 cols
NASA ASRS (3 sets)	Real	data/raw/nasa_asrs/	150 parsed reports
FAA Airworthiness Dir.	Real	data/raw/faa_ad/real/	500 metadata, 49 full-text
Synthetic Techlogs	Synthetic	data/raw/techlogs/	4,425 records, 70 tails
Knowledge Corpus	Mixed	data/raw/knowledge_corpus/	1,213 documents total
C-MAPSS-style Signals	Synthetic	data/raw/cmapss_signals/	3,915 windows
Triage Splits	Processed	data/processed/triage/	3,540 / 442 / 443
Recurrence Splits	Processed	data/processed/recurrence/	3,470 / 372 / 373
SDR Normalised Records	Processed	data/processed/	196,118 records
Defect-Action Pairs	Processed	data/processed/	26,210 pairs
QA Benchmark	Processed	data/processed/	189 items

Table 1: Dataset Summary

A few observations from working with this data are worth noting. The FAA SDR free text is genuinely noisy -- it arrives in upper case, uses heavy abbreviations, and often contains both the defect and the corrective action run together without a clean separator. The regex-based splitter handles most cases but approximately 10% of records were not cleanly separable and were excluded from the pairs file.

The 30,081 records in the OTHER taxonomy bucket represent a real limitation of the heuristic ATA mapping approach. Many cover ATA chapters outside the project's 15-chapter scope (e.g., ATA 25 cabin equipment, ATA 45 central maintenance). These are preserved with the mapping_method field flagged as "unmapped" for future ML-based recovery.

5. PROGRESS REPORT

This section summarises the actual progress made against the original dissertation plan, the work that remains, and the technical challenges that were encountered and resolved during the data preparation phase.

5.1 Milestone Status

No.	Phase	Planned Dates	Deliverable	Status
1	Dissertation Outline	05-10 May 2026	Literature review, problem statement, outline report	DONE
2	Data Acquisition & Prep.	10 May - 15 Jun 2026	All public datasets downloaded; synthetic corpus generated; all splits built	DONE
3	RAG Pipeline & Vector Store	16 Jun - 05 Jul 2026	Corpus embedded into Chroma; retrieval agent operational; Recall@k evaluated on QA benchmark	IN PROG.
4	Triage & Recurrence Models	06-20 Jul 2026	Multimodal MLP triage trained; LSTM recurrence trained; ablation baselines	PENDING
5	LLM Fine-tuning & Agent Integ.	21 Jul - 06 Aug 2026	LoRA fine-tune Mistral-7B; LangGraph orchestration; safety layer	PENDING
6	API, UI & Evaluation	07-13 Aug 2026	FastAPI backend; Streamlit UI; full evaluation	PENDING
7	Review & Submission	14-19 Aug 2026	Supervisor review; final corrections; submission	PENDING

Table 3: Milestone Progress Status as of June 2026

5.2 Summary of Completed Work

Two of the seven planned phases are fully complete at the mid-semester point, and the third (RAG pipeline) has just commenced. The following items have been delivered:

- Literature review covering RAG, LLMs, multimodal learning, predictive maintenance, agentic AI frameworks, and EASA AI governance (20 references reviewed).
- Full data acquisition pipeline in src/data/ -- downloads, normalises, and generates all datasets deterministically (seed=42) with one command.
- Real data: NASA C-MAPSS (FD001-FD004), FAA SDR 2023-2025 (~197k records), NASA ASRS (150 parsed reports), FAA AD (500 rules, 49 with full text).
- Synthetic data: 4,425 techlogs across 70 tails, 563 corpus documents, 3,915 signal windows, 189-item QA benchmark.
- Processed outputs: 196,118 normalised SDR records, 26,210 defect-to-action pairs, chronological train/val/test splits for triage and recurrence tasks.
- Knowledge corpus: 1,213 retrievable documents (synthetic + real FAA ADs + ASRS cases).

- Custom ASRS PDF parser (src/data/parse_asrs.py) extracting structured records from de-identified NASA report sets.

- Reproducible Python environment: venv with all required packages installed (PyTorch, Transformers, LangChain, LangGraph, Chroma, FastAPI, Streamlit, etc.).

5.3 Work Remaining

The following phases remain to be completed in the second half of the semester. The RAG pipeline is the immediate priority, as it underpins the retrieval agent, the QA benchmark evaluation, and part of the reasoning agent's context.

RAG Pipeline (current)

Embed all 1,21
vector store. In
nDCG on the 18

Triage Classifier

Train a mult
slope/last-value
metadata + sig
recurrence-posi

Recurrence LSTM

Train an LSTM
balanced. Evalu

LLM Fine-tuning

Apply LoRA
constraints, QL
task: given a de

Agent Orchestration & Safety Layer

Wire all agen
configurable via
logging (JSON

FastAPI Backend & Streamlit UI

Expose the age
defect input fo
recurrence risk

System Evaluation

Run the full
precision/recall
latency, audit co

5.4 Challenges Encountered and Overcome

The data preparation phase threw up several practical obstacles that were not obvious from the outset. Each one required diagnosis and a workaround before the pipeline could proceed.

Challenge 1: FAA Federal Register AD full-text access blocked

The Federal F
downloader trie
rather than the
AD content. Th
repository, whi
recovered full t
is a known limi

Challenge 2: SDR free-text parsing -- no clean defect/action separator

FAA SDR rec
"Discrepancy"
APPROACH. C
"C/A", some us
built that tests
26,210 clean d
normalised out
training data.

Challenge 3: ASRS PDF column-layout causing text block merging

The three NAS
from a scanned
producing garbl
as the ground-
cleanly. All 1
incomplete com

Challenge 4: ATA taxonomy coverage gap -- 30,081 SDR records unmapped

The project's de
a widebody ca
equipment (AT
others that fall
would introduc
and mapping_m
be trained on th

Challenge 5: Python environment not present on new development machine

Following a la
Windows mach
Store rather tha
environment w
LangGraph, Ch
was restored fr

against the expe

Challenge 6: NASA C-MAPSS nested ZIP extraction

The C-MAPSS
extract left the
directory tree, c
After this fix, a
correctly extrac

Challenge 7: Maintaining synthetic data internal consistency

The synthetic p
pairs across sep
document that
component. Ac
downstream ge
post-generation
563 synthetic co

6. FUTURE PLAN

No.	Phase	Start - End Date	Work to be Done	Status
1	Dissertation Outline	05 May - 10 May 2026	Literature review and Dissertation Outline	DONE
2	Data Acquisition & Prep	10 May - 15 Jun 2026	Datasets downloaded; synthetic corpus; all splits built	DONE
3	RAG Pipeline & Vector Store	16 Jun - 05 Jul 2026	Corpus embedded; retrieval agent; Recall@k and nDCG on QA	IN PROG.
4	Triage & Recurrence Models	06 Jul - 20 Jul 2026	MLP triage; LSTM recurrence; ablation studies	PENDING
5	LLM & Agent Integration	21 Jul - 06 Aug 2026	LoRA Mistral-7B; LangGraph; safety layer	PENDING
6	API, UI & Evaluation	07 Aug - 13 Aug 2026	FastAPI; Streamlit; full system evaluation	PENDING
7	Review & Submission	14 Aug - 19 Aug 2026	Supervisor review; final corrections & submission	PENDING

Table 4: Project Plan

7. ABBREVIATIONS

Abbreviation	Full Form
AD	Airworthiness Directive
AHM	Aircraft Health Monitoring
AMM	Aircraft Maintenance Manual
ASRS	Aviation Safety Reporting System (NASA)
ATA	Air Transport Association (chapter numbering standard)
AUPRC	Area Under Precision-Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BITE	Built-In Test Equipment
CDL	Configuration Deviation List
C-MAPSS	Commercial Modular Aero-Propulsion System Simulation
CMC	Central Maintenance Computer
EASA	European Union Aviation Safety Agency
EGT	Exhaust Gas Temperature
EO	Engineering Order
FAA	Federal Aviation Administration
GPU	Graphics Processing Unit
JASC	Joint Aircraft System/Component
LLM	Large Language Model
LoRA	Low-Rank Adaptation
LSTM	Long Short-Term Memory (recurrent neural network variant)
MEL	Minimum Equipment List
MLP	Multilayer Perceptron
MRO	Maintenance, Repair and Overhaul
nDCG	Normalised Discounted Cumulative Gain
NLP	Natural Language Processing
QA	Question Answering
RAG	Retrieval-Augmented Generation
RUL	Remaining Useful Life
SDR	Service Difficulty Report
SRM	Structural Repair Manual

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